

1.1 Vector Spaces

A **vector space** $\mathbb{V}$ is a set of vectors and two operators that satisfy the following properties:

1. Vector Addition

- Associative: $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$ for any $\mathbf{v}, \mathbf{u}, \mathbf{w} \in \mathbb{V}$.
- Commutative: $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ for any $\mathbf{v}, \mathbf{u} \in \mathbb{V}$.
- Additive Identity (Zero Vector): There exists an additive identity $\mathbf{0} \in \mathbb{V}$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$ for any $\mathbf{v} \in \mathbb{V}$.
- Additive Inverse (Negative): For any $\mathbf{v} \in \mathbb{V}$, there exists $-\mathbf{v} \in \mathbb{V}$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$. We call $-\mathbf{v}$ the additive inverse of $\mathbf{v}$.
- Closure under vector addition: For any two vectors $\mathbf{v}, \mathbf{u} \in \mathbb{V}$, their sum $\mathbf{v} + \mathbf{u}$ must also be in $\mathbb{V}$.

2. Scalar Multiplication

- Associative: $\alpha(\beta\mathbf{v}) = (\alpha\beta)\mathbf{v}$ for any $\mathbf{v} \in \mathbb{V}$, $\alpha, \beta \in \mathbb{R}$.
- Multiplicative Identity: There exists $1 \in \mathbb{R}$ where $1 \cdot \mathbf{v} = \mathbf{v}$ for any $\mathbf{v} \in \mathbb{V}$. We call 1 the multiplicative identity.
- Distributive in vector addition: $\alpha(\mathbf{u} + \mathbf{v}) = \alpha\mathbf{u} + \alpha\mathbf{v}$ for any $\alpha \in \mathbb{R}$ and $\mathbf{u}, \mathbf{v} \in \mathbb{V}$.
- Distributive in scalar addition: $(\alpha + \beta)\mathbf{v} = \alpha\mathbf{v} + \beta\mathbf{v}$ for any $\alpha, \beta \in \mathbb{R}$ and $\mathbf{v} \in \mathbb{V}$.
- Closure under scalar multiplication: For any vector $\mathbf{v} \in \mathbb{V}$ and scalar $\alpha \in \mathbb{R}$, the product $\alpha\mathbf{v}$ must also be in $\mathbb{V}$.

Note: Sometimes we will be dealing with complex scalars instead if our vectors are complex-valued. In that case, all scalars in the definition above will be in $\mathbb{C}$ instead of $\mathbb{R}$.

You have already seen vector spaces before! For example, $\mathbb{R}^n$ is the vector space of all n -dimensional vectors with real-valued components. With the operations of vector addition and scalar multiplication defined, you could show that $\mathbb{R}^n$ satisfies all the properties above. In fact, the set of all matrices of the same size $n \times m$ is also a vector space $\mathbb{R}^{n \times m}$ since it fulfills all of the properties above as well. In this class we will almost always deal with vector spaces containing vectors in $\mathbb{R}^n$ or $\mathbb{C}^n$, but may explore other vector spaces.

Additional Resources For more on vector spaces, read *Strang* pages 123 - 125 and try Problem Set 3.1.

In *Schaum's*, read pages 112-114 and try problems 4.1, 4.2, and 4.71 to 4.76. *Extra: Read and Understand Polynomial Spaces, Spaces of Arbitrary "Field."*

1.2 Subspace

In the previous section, we introduced the concept of a vector space. However, it is often useful to look at a specific subset of the vector space instead, which we will call the subspace.

Definition 1.1 (Subspace): A subspace $\mathbb{U}$ consists of a subset of the vector space $\mathbb{V}$ that satisfies the following three properties:

- Contains the zero vector: $\mathbf{0} \in \mathbb{U}$.
- Closed under vector addition: For any two vectors $\mathbf{v}_1, \mathbf{v}_2 \in \mathbb{U}$, their sum $\mathbf{v}_1 + \mathbf{v}_2$ must also be in $\mathbb{U}$.
- Closed under scalar multiplication: For any vector $\mathbf{v} \in \mathbb{U}$ and scalar $\alpha \in \mathbb{R}$ (or $\mathbb{C}$), the product $\alpha\mathbf{v}$ must also be in $\mathbb{U}$.

Some texts replace the first condition in this definition with the condition that $\mathbb{U}$ must be nonempty. Both conditions imply each other as if $\mathbb{U}$ contains $\mathbf{0}$, it is nonempty. If $\mathbb{U}$ is nonempty and thus contains a vector $\mathbf{v}$, then using the third condition, we have that $0\mathbf{v} = \mathbf{0}$. But the best way to test for a subspace is to look first for the zero vector. If $\mathbf{0}$ is in $\mathbb{U}$, then the other 2 conditions must be checked. If $\mathbf{0}$ is not in $\mathbb{U}$, then $\mathbb{U}$ cannot be a subspace and the other conditions do not need to be checked.

In the following sections, we will explore a few key vector spaces and subspaces.

Additional Resources For more on subspaces, read *Strang* pages 125 - 127 and try Problem Set 3.1. In *Schaum's*, read pages 117-119 and try Problems 4.8 to 4.12, and 4.77 to 4.82.

1.3 Examples of Vector Spaces

1.3.1 Cartesian Spaces

The standard geometric spaces (marked by perpendicular axes like x and y) are the **Cartesian spaces**, which can come in both real and complex forms.

To be more specific, the n -dimensional real Cartesian space is notated as $\mathbb{R}^n$ and a vector $\mathbf{v} \in \mathbb{R}^n$ is usually represented as a set of n real numbers as follows:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \quad (1)$$

where $v_1, v_2, \dots, v_n \in \mathbb{R}$.

Formally, this vector space can be represented with:

$$\mathbb{R}^n = \left\{ \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \mid v_1, v_2, \dots, v_n \in \mathbb{R} \right\} \quad (2)$$

An example of a vector in $\mathbb{R}^3$ would be:

$$\begin{bmatrix} 1 \\ -\pi \\ e^4 \end{bmatrix} \in \mathbb{R}^3 \quad (3)$$

The n -dimensional complex Cartesian space is notated as $\mathbb{C}^n$ and a vector $\mathbf{u} \in \mathbb{C}^n$ is usually represented as a set of n complex numbers as follows:

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad (4)$$

where $u_1, u_2, \dots, u_n \in \mathbb{C}$.

Formally, this vector space can be represented with:

$$\mathbb{C}^n = \left\{ \mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \mid u_1, u_2, \dots, u_n \in \mathbb{C} \right\} \quad (5)$$

An example of a vector in $\mathbb{C}^4$ would be:

$$\begin{bmatrix} i \\ 2 \\ 3 - 4i \\ 5 + 6i \end{bmatrix} \in \mathbb{C}^4 \quad (6)$$

Note that complex Cartesian spaces can be defined in terms of real Cartesian spaces similar to how we do in the

scalar case:

$$\mathbf{u} = \mathbf{x} + i\mathbf{y} \in \mathbb{C}^n \quad (7)$$

with $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

For our example vector, this format looks as follows:

$$\begin{bmatrix} i \\ 2 \\ 3-4i \\ 5+6i \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 3 \\ 5 \end{bmatrix} + i \begin{bmatrix} 1 \\ 0 \\ -4 \\ 6 \end{bmatrix} \quad (8)$$

Furthermore, if we set $\mathbf{y} = \mathbf{0}$ in this form, we obtain the real Cartesian space $\mathbb{R}^n$. You can convince yourself that $\mathbb{R}^n$ is a valid subspace of $\mathbb{C}^n$.

1.3.2 Polynomial Spaces

Consider a standard n th degree polynomial:

$$p(x) = p_0 + p_1x + p_2x^2 + \dots + p_nx^n \quad (9)$$

Notice that what makes polynomials distinct from other polynomials of the same degree are simply the coefficients of each power term ($x^0 = 1, x^1 = x, x^2, \dots, x^n$). There are exactly $n + 1$ coefficients for an n th degree polynomial. Thus, polynomials can be represented as vector spaces where the components of the vector are the coefficients:

$$\mathbf{p} = \begin{bmatrix} p_0 \\ p_1 \\ \vdots \\ p_n \end{bmatrix} \quad (10)$$

For example, consider the polynomial:

$$p(x) = 5 + \pi x - 3.2x^2 \quad (11)$$

By simply identifying the coefficients, the vector representation would be:

$$\mathbf{p} = \begin{bmatrix} 5 \\ \pi \\ -3.2 \end{bmatrix} \quad (12)$$

There is a one-to-one mapping from addition and subtraction of polynomials in this vector representation to addition and subtraction of polynomials in the standard function representation.

1.4 Span

It can be confusing to distinguish a vector space from the span of a set of vectors. For example, what does it mean for a set of vectors to live in $\mathbb{R}^3$ (a 3D volume), but *span* only a 2D plane?

Intuitively, a **vector space** defines the physical ambient space that your vectors live in, while **span** describes the specific region a collection of vectors can occupy ('degrees of freedom' within your vector space). That's how a collection of k n -dimensional vectors lives in a $\mathbb{R}^n$ space, but can only 'span' at most a k -dimensional subset of that.

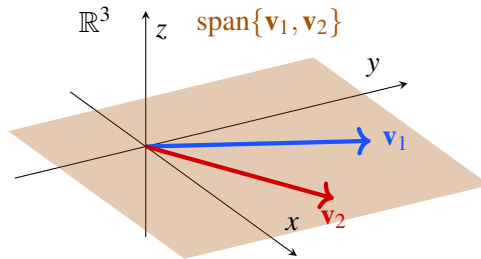


Figure 1: $\mathbf{v}_1$ and $\mathbf{v}_2$ span a 2D plane inside $\mathbb{R}^3$.

Definition 1.2 (Span): The span of a set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is the set of all of their linear combinations:

$$\text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_k\}) = \{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_k\mathbf{v}_k \mid c_1, \dots, c_k \in \mathbb{R}\}. \quad (13)$$

The span of a set of vectors is always a subspace of the vector space they live in (see Section 1.2): it contains the zero vector (set every $c_i = 0$), and it is closed under addition and scalar multiplication, since combining linear combinations just gives you another linear combination.

1.4.1 Proving Two Spans Are Equal

A span is just the set of vectors you can reach by combining a collection of vectors, so two spans are equal when they let you reach exactly the same vectors. Concretely, $\text{span}(S_1) = \text{span}(S_2)$ means every vector you can build from S_1 can also be built from S_2 , *and* every vector you can build from S_2 can also be built from S_1 .

Concretely, for $S_1 = \{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ and $S_2 = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$, proving $\text{span}(S_1) = \text{span}(S_2)$ comes down to two steps:

- **Step 1:** Show that every vector in S_1 can be written as a linear combination of S_2 (written $\text{span}(S_1) \subseteq \text{span}(S_2)$):

$$\mathbf{u}_i = c_{i1}\mathbf{v}_1 + c_{i2}\mathbf{v}_2 + \dots + c_{im}\mathbf{v}_m. \quad (14)$$

- **Step 2:** Show the reverse: every vector in S_2 can be written as a linear combination of S_1 (written $\text{span}(S_2) \subseteq \text{span}(S_1)$):

$$\mathbf{v}_j = d_{j1}\mathbf{u}_1 + d_{j2}\mathbf{u}_2 + \dots + d_{jk}\mathbf{u}_k. \quad (15)$$

Once both directions check out, $\text{span}(S_1) = \text{span}(S_2)$.

Example 1.1 (Proving Two Spans Are Equal: A Concrete Case):

Prove $\text{span}\left(\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right\}\right) = \text{span}\left(\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}\right)$.

In this case, since we are already given the two vectors, we can directly check the spanning vectors themselves. This is equivalent to proving the above two steps, since any other vector in a span is just a linear combination of these.

Step 1: We show $\text{span}\left(\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right\}\right) \subseteq \text{span}\left(\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}\right)$ by writing each vector in the first set as a linear combination of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$:

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Step 2: We show the reverse inclusion by writing $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ as linear combinations of $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} = -\frac{1}{3} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \frac{2}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Since both inclusions hold, $\text{span}\left(\left\{\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \end{bmatrix}\right\}\right) = \text{span}\left(\left\{\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix}\right\}\right)$, as desired! In particular, notice that both sets span all of $\mathbb{R}^2$.

1.5 Bases

We can use a series of vectors to define a vector space. We call this set of vectors a **basis**, which we define formally below:

Definition 1.3 (Basis): Given a vector space $\mathbb{V}$, a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a **basis** of the vector space if it satisfies the following two properties:

- $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent vectors
- $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ span the vector space

A set of vectors is linearly independent if and only if the only scalars $\alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{R}$ that make $0 = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$

$\dots + \alpha_n \mathbf{v}_n$ hold true are $\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$. In other words, the only way to write the zero vector as a linear combination of the vectors is by setting all of the coefficients to 0.

A set of vectors spans a vector space if and only if for any vector $\mathbf{v} \in \mathbb{V}$, there exist scalars $\alpha_1, \alpha_2, \dots, \alpha_n \in \mathbb{R}$ such that $\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$.

Intuitively, a basis of a vector space is the *minimum* set of vectors needed to represent all vectors in the vector space. If a set of vectors is linearly dependent and “spans” the vector space, it is still not a basis because we can remove at least one vector from the set and the resulting set will still span the vector space.

The next natural question to ask is: Given a vector space, is the basis unique? Intuitively, it is not because multiplying one of the vectors in a given basis by a nonzero scalar will not affect the linear independence or span of the vectors. We could alternatively construct another basis by replacing one of the vectors with the sum of itself and any other vector in the set.

To illustrate this mathematically, suppose $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a basis for the vector space we are considering. Then

$$\{\alpha \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \quad (16)$$

where $\alpha \neq 0$ is also a basis because multiplying a vector by a nonzero scalar also does not change the linear independence of the set of vectors. In addition, we know that

$$\text{span}(\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}) = \text{span}(\{\alpha \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}). \quad (17)$$

because any vector in $\text{span}(\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\})$ can be created as a linear combination of the set $\{\alpha \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ by dividing the scale factor on $\mathbf{v}_1$ by α . We can use a similar argument to show that $\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is also a basis for the same vector space.

Example 1.2 (Vector space $\mathbb{R}^3$): Let's try to find a basis for the vector space $\mathbb{R}^3$. We want to find a set of vectors that can represent any vector of the form $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ where $a, b, c \in \mathbb{R}$. One basis could be the set of standard unit vectors:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

The set of vectors is linearly independent and we can represent any vector $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ in the vector space using the three vectors:

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}. \quad (18)$$

Alternatively, we could show that

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

is a basis for the vector space.

Now that we have defined bases, we can define the dimension of a vector space.

Definition 1.4 (Dimension): The dimension of a vector space is the number of vectors needed to construct a basis.

Since each basis vector can be scaled by one coefficient, the dimension of a space is the fewest number of parameters needed to describe an element or member of that space. The dimension can also be thought of as the degrees of freedom of your space – that is, the number of parameters that can be varied when describing a member of that space.

Example 1.3 (Dimension of $\mathbb{R}^3$): Previously, we identified a basis

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

for the vector space $\mathbb{R}^3$. The basis consists of three vectors, so the dimension of the vector space is three.

Note that a vector space can have many bases, but each basis must have the same number of vectors. We will not prove this rigorously, but let's illustrate our arguments. Suppose a basis for the vector space we're considering has n vectors. This means that the minimum number of vectors we can use to represent all vectors in the vector space

is n , because the vectors in the basis would not be linearly independent if the vector space could be represented with fewer vectors. Then we can show that any set with less than n vectors cannot be a basis because it does not have enough vectors to span the vector space — there would be some vectors in the vector space that cannot be expressed as a linear combination of the vectors in the set. In addition, we can show that any set with more than n vectors must be linearly dependent and therefore cannot be a basis. Combining the two arguments, we have that any other set of vectors that forms a basis for the vector space must have exactly n vectors.

Just as basis and dimension are defined for vector spaces, they have equivalent definitions for subspaces. A basis of a subspace is a set of linearly independent vectors that span the subspace, and the dimension of a subspace is the number of vectors in its basis.

Note: A set of k vectors in $\mathbb{R}^n$ can span a subspace of dimension at most $\min(k, n)$; you can't express more independent directions than the vectors you started with, and you cannot exceed the dimension of the ambient space either.

Example 1.4 (n Independent Vectors in an n -Dimensional Subspace Must Span It):

Claim: If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a set of n linearly independent vectors in an n -dimensional subspace U , then $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ spans U and is a basis for U .

We prove this by contradiction. Suppose $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ does not span U , so there is some $\mathbf{u} \in U$ with $\mathbf{u} \notin \text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_n\})$. We show this forces $\{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{u}\}$ to be linearly independent. Suppose

$$c_1\mathbf{v}_1 + \dots + c_n\mathbf{v}_n + c_{n+1}\mathbf{u} = \mathbf{0}. \quad (19)$$

Case 1 ($c_{n+1} \neq 0$): We could solve for

$$\mathbf{u} = -\frac{c_1}{c_{n+1}}\mathbf{v}_1 - \dots - \frac{c_n}{c_{n+1}}\mathbf{v}_n, \quad (20)$$

contradicting $\mathbf{u} \notin \text{span}(\{\mathbf{v}_1, \dots, \mathbf{v}_n\})$.

Case 2 ($c_{n+1} = 0$): This leaves

$$c_1\mathbf{v}_1 + \dots + c_n\mathbf{v}_n = \mathbf{0}; \quad (21)$$

since $\mathbf{v}_1, \dots, \mathbf{v}_n$ are independent, every $c_i = 0$ too. So $\{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{u}\}$ is a set of $n + 1$ linearly independent vectors in U . But U is n -dimensional, and we argued above that no more than n vectors in an n -dimensional space can be independent — a contradiction. So $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ must span U , and being independent, it is indeed a basis for U .

Additional Resources For more on bases, read *Strang* pages 167 - 171 and try Problem Set 3.4. *Extra:* Read Sections on Matrix and Function Space.

In *Schaum's*, read pages 124-126 and pages 127-129. Try Problems 4.24 to 4.28, 4.97 to 4.103, and 4.33 to 4.40.